DO NOW — RECALL

Lesson 106 · What an Operating System Does

Name: _____ Class: _____ Date: _____

1. Distinguish RAM from virtual storage.

2. In a BST, where is a value less than the current node placed?

3. Why can an array provide direct indexed access?

4. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

RAM is local working memory; virtual storage presents remote/pooled file storage.

2. Model answer

In its left subtree (under the chosen no-duplicate convention).

3. Model answer

Its element locations can be calculated from base address and index.

4. Model answer

Sum 0 and carry-out 1.

By the end, you can…

• The OS creates and schedules processes; programs share CPU time.

• Memory, files and device access are mediated by the OS.

• Interfaces, user accounts and access controls make a computer usable.

Vocabulary — know these terms

operating system — System software managing hardware resources and services.

process — An executing instance of a program.

device driver — Software allowing the OS to communicate with hardware.

spooling — Queuing data such as print jobs for a slower peripheral.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Identify concurrent processes; explain the print queue/device driver; describe file and network access. Printing and downloading share resources without either application controlling the hardware directly.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Manage processes

The OS creates and schedules processes; programs share CPU time.

Manage resources

Memory, files and device access are mediated by the OS.

Provide services

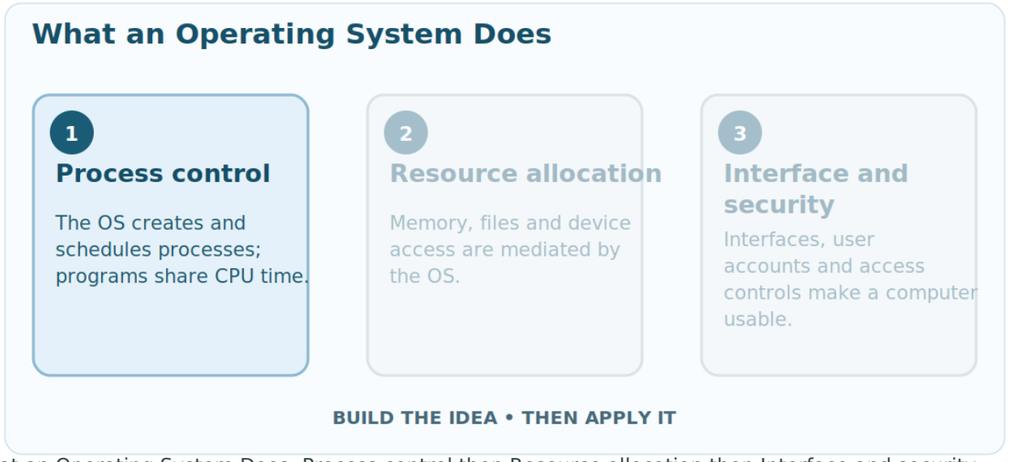
Interfaces, user accounts and access controls make a computer usable.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

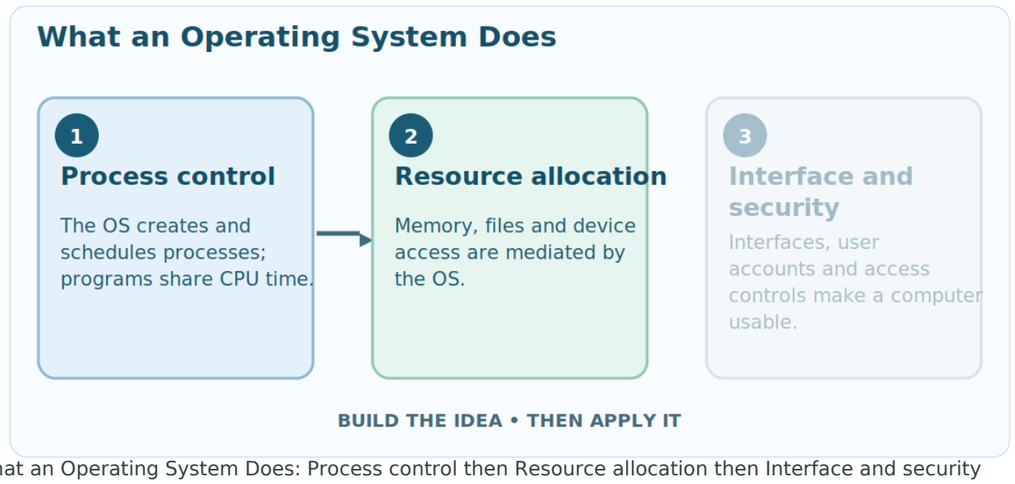
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



What an Operating System Does: Process control then Resource allocation then Interface and security

New Learning



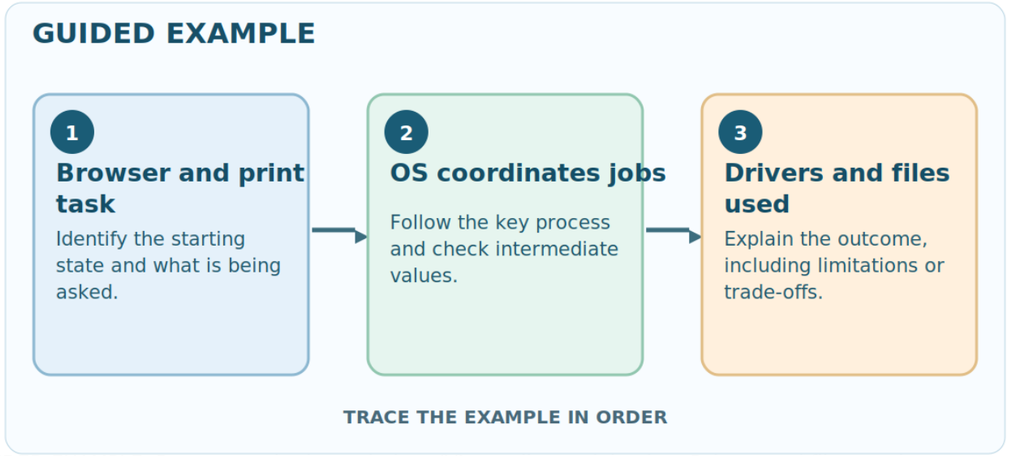
What an Operating System Does: Process control then Resource allocation then Interface and security

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil prints while a browser downloads a file. Which OS responsibilities are involved?

Identify the inputs, relevant facts and exact output or judgement required.



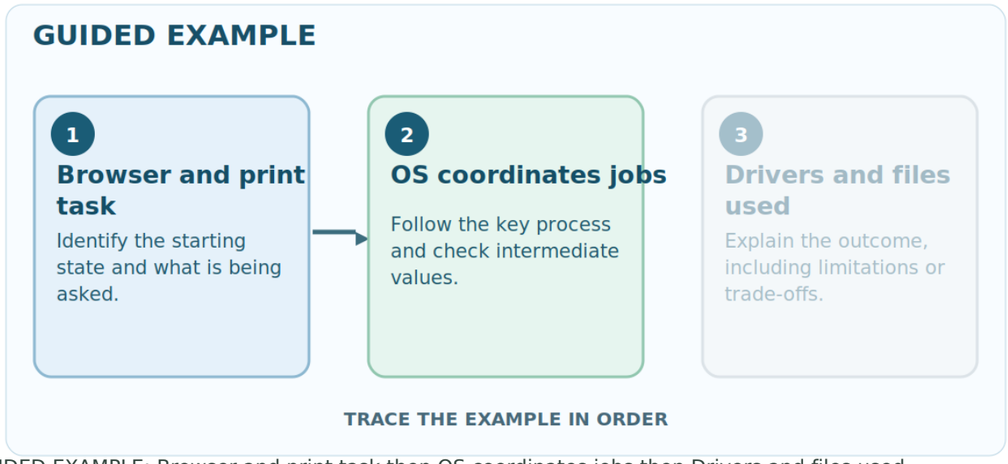
GUIDED EXAMPLE: Browser and print task then OS coordinates jobs then Drivers and files used

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil prints while a browser downloads a file. Which OS responsibilities are involved?

Identify the inputs, relevant facts and exact output or judgement required.

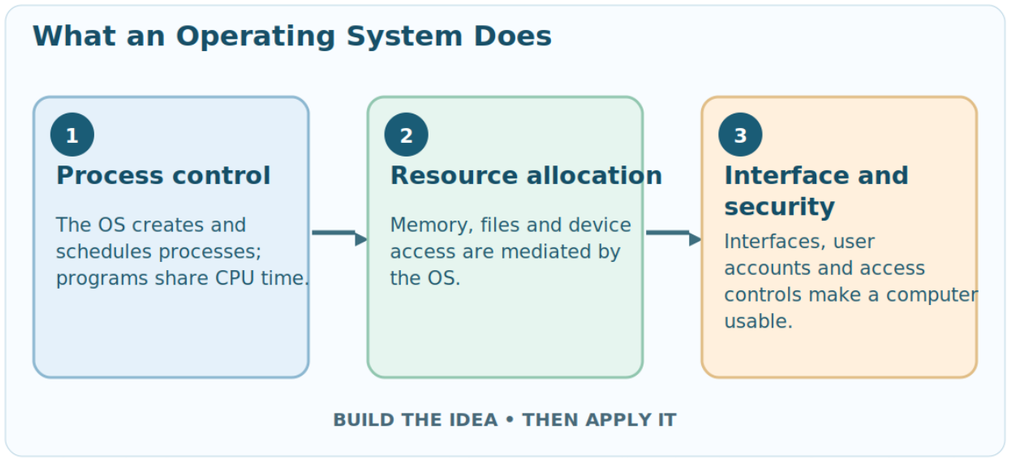


GUIDED EXAMPLE: Browser and print task then OS coordinates jobs then Drivers and files used

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



What an Operating System Does: Process control then Resource allocation then Interface and security

I DO — WORKED EXAMPLE

Follow the worked method

A word processor saves a document while music is playing. Explain three OS services involved.

Model answer

The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected.

Check: link each claim to the scenario and use precise OCR terminology.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The operating system only provides a graphical user interface.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the claim that application programs can directly manage all hardware without system software. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A word processor saves a document while music is playing. Explain three OS services involved.

The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected.



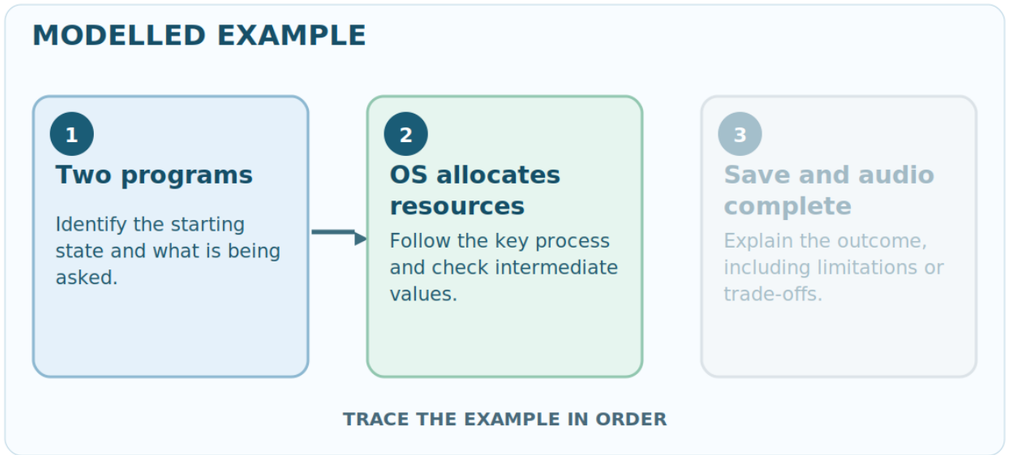
MODELLED EXAMPLE: Two programs then OS allocates resources then Save and audio complete

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A word processor saves a document while music is playing. Explain three OS services involved.

The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected.



MODELLED EXAMPLE: Two programs then OS allocates resources then Save and audio complete

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe three functions of an operating system, linking each to a user action. [6]

Question 2

A computer has three applications requesting the printer. Explain how the OS can handle this. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

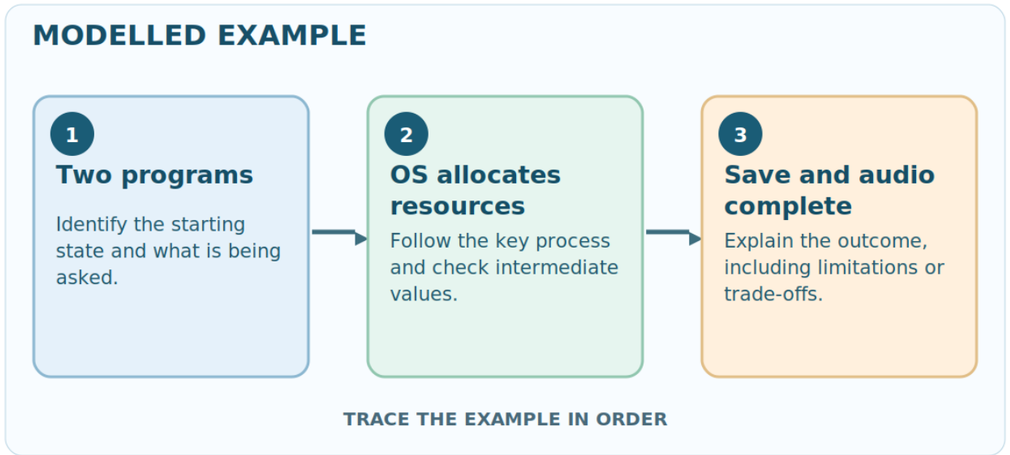
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A word processor saves a document while music is playing. Explain three OS services involved.

The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected.



MODELLED EXAMPLE: Two programs then OS allocates resources then Save and audio complete

WE DO — GUIDED PRACTICE

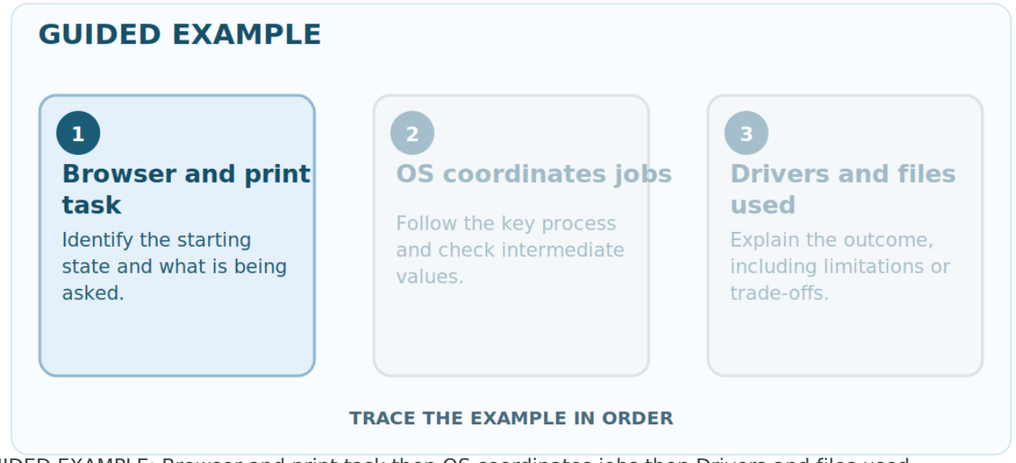
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil prints while a browser downloads a file. Which OS responsibilities are involved?

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Browser and print task then OS coordinates jobs then Drivers and files used